Emil Kasper

Vienna and Zurich kaspere@ethz.ch emil-kasper

Profile

Computer Science student at ETH Zurich, with a focus on reinforcement learning, robot learning, and human-AI interaction. Experience spanning machine-learning research, software engineering, and robotics.

Experience

Agile Robots SE, Machine Learning Engineer (Working Student) — Munich, Germany
- Designing and running machine-learning training environments. — Oct 2026 – present
- Collecting teleop data and preparing the data for model training. — 1 month

IVIA Lab, ETH Zurich, Research Fellow (ETH SSRF) — Zurich, Switzerland
Supervised by Prof. Menna El-Assady. — July 2026 – Aug 2026 — 2 months
- Selected as one of 20 students from over 4,000 applicants for the ETH Student Summer Research Fellowship.
- Working on agentic AI and agent-agent communication.

Siemens, Software Engineer (Working Student) — Vienna, Austria
- Contributing to a test simulation framework for the European Train Control System (ETCS). — July 2025 – June 2026
- Implementing the autonomous train operation (ATO) mode in C# for test simulations. — 1 year

Institute for Visual Computing and Human-Centered Technology, TU Wien, Teaching Assistant — Vienna, Austria
Course: Interface and Interaction Design. — Mar 2025 – June 2026 — 1 year 4 months
- Guiding students through the lifecycle of user-centered product development.
- Providing feedback and grading for 45 students.

Machine Learning Research Unit, TU Wien, Undergraduate Researcher — Vienna, Austria
Supervised by Prof. Clemens Heitzinger. — Aug 2025 – Apr 2026 — 9 months
- Investigating the convergence and stability of Q-Learning and Speedy Q-Learning.
- Experimenting with PPO for robotic hand manipulation.

Education

MSc — **ETH Zurich**, Computer Science — Zurich, Switzerland
- Major in Machine Intelligence. — Sept 2026 – June 2028

BSc (Honors) — **TU Wien (Vienna University of Technology)**, Computer Science — Vienna, Austria
- Four-year honors degree with a minor in machine learning. — Oct 2022 – May 2026
- Graduated with distinction.
- Received merit-based scholarships.

Research

Asking Your Advisor: Learning When to Consult Stronger Models in Agentic Systems
Submitted to ICLR 2027. We train a meta-policy that learns when to consult a stronger advisor model and when to continue with a cheaper one, reaching a better success-cost Pareto frontier than standard advisor-executor setups. Preprint on arXiv.

Student Engagements

TU Wien Robotics Club, Machine Learning Engineer — Vienna, Austria
Apr 2026 – present
- Implementing VLAs for robotic applications with an SO101 arm.
- Building a chess robot that autonomously plays chess and moves the pieces.

European Student Robotics Association (ESRA), Executive Board Member, Partnerships — Europe
Sept 2026 – present
- Serving on the executive board of ESRA, the European Student Robotics Association.
- Driving partnerships to push the boundaries of European robotics.

TU Wien Space Team, Frontend Developer — Vienna, Austria
Mar 2025 – Oct 2025
- Contributing to the "Lamarr" project at the 2025 European Rocketry Challenge (EuRoC).
- Building frontend tooling for rocket system control.

Programs

TU Wien Informatics, Bachelor with Honors — Vienna, Austria
Oct 2024 – May 2026
- Selected for the Informatics excellence program, a performance-based honors track comparable to the Anglo-American honors model.
- Performance-based admission and graduation places graduates among the top 5 percent of students at TU Wien.
- Extends the studies by one year with research-oriented and master's-level courses (program website).

European Student Assembly 2026, EUC Voices, Delegate — Strasbourg, France
Jan 2026 – Apr 2026
- Representing the EULiST alliance at the European Student Assembly 2026.
- Debating key issues and advocating for policy changes at the European level.
- Writing policy recommendations on empowering inclusive digital literacy and building resilience.

Skills

Programming
Python, Java, C#, HTML, CSS, JavaScript

Machine Learning
Model development and evaluation

Reinforcement Learning
Algorithms, use cases, and applications

Robot Learning
Theory of VLAs, applied ML and implementation of VLAs, applying reinforcement learning for robotics tasks

Software Engineering
Git, Docker, Maven, REST APIs

Human-AI collaboration
shared-initiative interaction, human-in-the-loop

Generative AI
Agentic AI, prompting, API-based integration, prototype development

Languages

German
Native

English
Fluent

Russian
Fluent